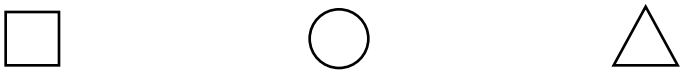
Use one unchanged bag containing both colors. Choose each draw independently without looking; each counter has the same chance. Return it and mix after every draw. A rule sees only the color word, in order.

Problem 1: Make a three-draw rule that gives a square, circle, or triangle with equal chances for every fixed bag containing both colors. Skip RRR and BBB; use every other word.



R R R	output	R R B	output
R B R	output	R B B	output
B R R	output	B R B	output
B B R	output	B B B	output

For each pair: BR gives a square, RB a circle, and RR or BB gives nothing. Make four draws from one fixed bag; use the first pair and the second pair. Keep both outputs already made.

BRRB → **BR | RB** → □ ○

four draws | first pair | second pair | both outputs kept

Problem 2: If both pairs were skipped, can you use them to make one extra fair output? Design a rule. Compare output totals on all sixteen equally likely stories from a one-red, one-blue bag. Would it also work for three reds and one blue?

RR | RR outputs:	**RR | RB** outputs:	**RR | BR** outputs:	**RR | BB** outputs:
RB | RR outputs:	**RB | RB** outputs:	**RB | BR** outputs:	**RB | BB** outputs:
BR | RR outputs:	**BR | RB** outputs:	**BR | BR** outputs:	**BR | BB** outputs:
BB | RR outputs:	**BB | RB** outputs:	**BB | BR** outputs:	**BB | BB** outputs:

	□	○
Basic		
Recycled		

Problem 3: Make two cups of four equal-chance tickets, each showing a left shape and a right shape. At each position, square and circle must have equal chances. One cup must give only two possible pairs; the other must give all four. Does fairness at each position make all four shape pairs equally likely?

Cup 1

left	right

left	right

left	right

left	right

Cup 2

left	right

left	right

left	right

left	right